

Vivek Ranjan

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Staff Software Engineer with 10+ years track record of leading agile teams and delivering high-impact, scalable, full-stack systems using Python, Java, TypeScript, React, Next.js, and AWS. Enjoys collaborating across teams to solve significant, real-world problems while relentlessly discovering ways to optimize software systems and organizational delivery processes.

SKILLS

BACKEND

Python, NodeJS, Java, SQL (MS SQL Server, Postgres)

FRONTEND

Typescript, React, Next.js

INFRASTRUCTURE/OTHER

Agentic AI Coding (Cursor, Claude Code, Junie, etc), AWS (Lambda, API Gateway, SQS/SNS, ECS, ECR, Fargate, ElasticBeanstalk, Cloudformation, CDK, S3, EC2, RDS, IAM, etc), Datadog, Grafana (Loki, Prometheus, etc), CI/CD pipelines, Docker, Kubernetes (kubectl, Helm, k9s, etc), Agile, Scrum, Shell

WORK EXPERIENCE

Staff Software Engineer

Capsule

April 2025 - Present

- Designed and implemented an LLM-powered AI agent system using Pydantic AI that accelerated prior authorization processing by 70%, directly improving patient access to medication.
- Designed event-driven task routing system using microservices to distribute prescription processing failures to pharmacists and technicians based on real-time availability
- Improved database performance leading to 21% reduction in CPU usage, 35% reduction in IOPS and 50% reduction in costs.
- Backend - Python, pytest, Django, PostgreSQL
- Frontend - Typescript, Next.js, React, Vite
- Infrastructure/Other - Docker, Kubernetes, AWS (Lambda, ElastiCache, EKS, RDS, SQS/SNS), Shell, CI/CD pipelines, Sentry, Grafana (Prometheus, Loki)

Staff Software Engineer

Renaissance Learning

February 2022 - April 2025

- Proposed & led project: Achieved 73% cost reduction by moving batch jobs to AWS Fargate.
- Created an IaC framework (Python, AWS CDK, Docker) that raised deployment frequency by 40% and streamlined batch-job development for all teams.
- Mentored multiple engineers leading to their promotions and improvement of their technical skills.
- Backend - Python, pytest, Java, Spring (MVC), Groovy/Spock, Microsoft SQL Server
- Frontend - Vue/Vuex, HTML/CSS, JavaScript, Jest
- Infrastructure/Other - Docker, AWS (ECS/ECR, Fargate, CDK, RDS, Aurora Serverless), Shell, CI/CD pipelines, Datadog, Cronitor

Lead Software Engineer

Renaissance Learning

June 2019 - February 2022

- Led project: UX redesign implementation of crucial progress monitoring tool used by educators and millions of students. Modernized frontend (Vue, Vuex, Jest) and optimized backend (Java, Spring, REST, Hibernate, MS SQL Server, Groovy/Spock).

- Proposed & led project: Reduced engineering effort by 4 weeks annually by building a microservice driven content-management system on AWS Aurora Serverless, Lambda, and API Gateway with a Vue front-end.
 - Proposed & led project: Doubled deployment frequency and saved 10 hrs/week by automating monolith deployments and adding end-to-end CI/CD tests with Docker and CloudFormation
 - Backend - Python, pytest, Java, Spring (MVC), Groovy/Spock, Microsoft SQL Server, AWS Aurora (serverless Postgres)
 - Frontend - Vue/Vuex, HTML/CSS, JavaScript, Jest
 - Infrastructure/Other - Docker, AWS (Lambda, API Gateway, ElasticBeanstalk, CloudFormation, IAM, RDS, Aurora Serverless), Shell, CI/CD pipelines, Datadog
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Lead Software Engineer

FastBridge Learning

February 2018 - June 2019

- Led project to deliver a web application that provides easy access to millions of data points that help educators understand the performance of all students in a mid-western state.
 - Worked with a scrum team to build a datawarehouse in AWS Redshift, a microservice serving REST APIs using Spring Boot, and a JS-based frontend to render data as user-friendly reports.
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Senior Software Engineer

FastBridge Learning

February 2017 - February 2018

- Reduced load time by 98% for a web-based graphical and tabular report used by educators to inform instructional methodology for at-risk students.
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Founding Engineer

FastBridge Learning

June 2015 - February 2017

- 4th engineer at Seed-stage startup that reached \$20M ARR and successful exit.

EDUCATION

Master of Science, Computer Science

University of Minnesota

2013 - 2016